Investigating the Association Between TP53 Mutation Types and Overall Survival in Lung Adenocarcinoma Using TCGA Data

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Abstract

Background: TP53 is the most frequently mutated tumour-suppressor gene in lung adenocarcinoma (LUAD), but "TP53-mutant" covers very different changes to the p53 protein. This project asked whether different types of TP53 mutation are associated with different overall survival (OS), and whether any association is independent of tumour stage.

Methods: Mutation and clinical data for 566 patients in the TCGA LUAD PanCancer Atlas study were downloaded from cBioPortal and processed in Python. TP53 mutations were classified as missense, truncating/splice (nonsense, frameshift, splice site), or other. OS was compared using Kaplan–Meier curves and log-rank tests, and using Cox proportional-hazards models adjusted for AJCC pathologic stage.

Results: 295 of 566 patients (52.1%) carried a protein-altering TP53 mutation; 61% of mutation records were missense, concentrated in the DNA-binding domain. Among 505 patients with survival data (182 deaths), TP53-mutant patients had shorter OS than wildtype patients (median 42.3 vs 53.7 months; log-rank $p = 0.033$), but survival did not differ significantly between the wildtype, missense, and truncating/splice groups ($p = 0.170$). Tumour stage was by far the strongest predictor (hazard ratios 2.5–3.9 versus stage I). After adjusting for stage, the overall TP53 effect weakened (HR 1.21, 95% CI 0.90–1.62), the missense effect disappeared (HR 0.99), and only truncating/splice mutations remained associated with worse OS (HR 1.52, 95% CI 1.01–2.29; $p = 0.043$).

Conclusion: The hypothesis that mutation type matters received only weak, borderline support. Much of the apparent TP53 effect overlapped with tumour stage, and the study had limited statistical power (about 48% to detect a hazard ratio of 1.5). Larger cohorts and finer, function-based mutation classes are needed to answer the question properly.

1. Introduction

1.1 Research question

Is the type of TP53 mutation in a patient's lung adenocarcinoma associated with how long that patient survives, and does any such association remain once tumour stage is taken into account?

1.2 TP53

Lung cancer is the leading cause of cancer death worldwide, and lung adenocarcinoma (LUAD) is its most common histological subtype. The TP53 gene encodes p53, a transcription factor often described as the "guardian of the genome". When a cell's DNA is damaged, p53 switches on genes that pause the cell cycle, repair the damage, or, if the damage is too severe, trigger programmed cell death (apoptosis) [7, 9]. Losing the p53 function lets damaged cells keep dividing and accumulate further mutations. This is why TP53 is the most frequently mutated gene across human cancers [10, 24] and is altered in roughly half of lung adenocarcinomas [1]. Whether TP53 status also predicts survival in lung cancer is less clear, and published studies have reached different conclusions [25].

1.3 Mutation Type

I began with a simple question: do TP53 mutations make lung cancer patients fare worse? While exploring the TCGA data, however, I realised that "TP53-mutant" is not a single biological category. A missense mutation swaps one amino acid and leaves a full-length protein. When it hits a "hotspot" in the DNA-binding domain, the mutant protein can block any remaining normal p53 (a dominant-negative effect) and may even gain new cancer-promoting functions [8]. A nonsense, frameshift, or splice-site mutation, by contrast, usually produces a shortened protein or none at all. In head and neck cancer, separating "disruptive" from "non-disruptive" TP53 mutations revealed survival differences that a simple mutant-versus-wildtype split had missed [11]. If mutation types have different biological consequences, they may also have different clinical consequences.

1.4 Hypothesis

Main hypothesis: different types of TP53 mutations are associated with different overall survival in patients with lung adenocarcinoma.

Specific prediction: patients with truncating/splice mutations, which most completely remove p53, would have the worst survival; wild-type patients the best; and missense patients somewhere in between.

Secondary question: is the association between TP53 mutation type and survival independent of tumour stage? Stage is the single most important prognostic factor in lung cancer [13], so an apparent TP53 effect could simply reflect TP53-mutant tumours being diagnosed at more advanced stages.

2. Methods

2.1 Data source

A single dataset was used throughout, so that the data source stayed clear and different versions were never mixed: Lung Adenocarcinoma (TCGA, PanCancer Atlas), study ID luad_tcga_pan_can_atlas_2018, downloaded as a ZIP archive from the cBioPortal Datasets page [4, 5]. I chose the PanCancer Atlas version over the Firehose Legacy and GDC versions because its mutation calls (the MC3 project [6]) and clinical endpoints (the TCGA Clinical Data Resource, TCGA-CDR [3]) were harmonised across all TCGA cancer types [2], with survival endpoints curated specifically for outcome studies. Two files were used: `data_mutations.txt` (one row per mutation, at the tumour-sample level) and `data_clinical_patient.txt` (one row per patient, including OS_MONTHS, OS_STATUS, AGE, SEX and AJCC pathologic stage). All data are public and de-identified.

Tumour stage: The first version of my Phase 1 table kept only survival, age and sex. For the stage-adjusted analysis, AJCC pathologic stage was added from the TCGA-CDR supplementary table [3], which is the source of cBioPortal's PanCancer Atlas clinical data (the same field appears as AJCC_PATHOLOGIC_TUMOR_STAGE in the clinical file, and the updated pipeline reads it directly). To confirm that the two sources matched, I compared them patient by patient. For all 505 patients with survival data, survival times were identical after converting days to months, and death status and sex agreed in 100% of cases.
Sub-stages were combined into stages I, II, III and IV (for example, IA and IB became I).

Eight patients whose stage was missing or marked "[Discrepancy]" were excluded from stage-adjusted models only.

2.2 Building the analysis table

All processing was done in Python 3 with pandas. The steps were:

1. **Load both files** → Lines starting with "#" are metadata and were skipped. I did not open the mutation file in Excel, which can silently mangle large tab-separated files.
2. **Extract TP53 records** (`Hugo_Symbol == "TP53"`) → This gave 318 mutation records from 295 tumour samples.
3. **Records are not patients** → One patient can carry several TP53 mutations, so the 318 records had to be collapsed into one row per patient before any survival analysis.
4. **Link samples to patients** → Mutation data use sample barcodes (e.g. TCGA-05-4382-01), whereas clinical data use patient IDs (TCGA-05-4382). The patient ID is the first three parts of the barcode. Every sample was a primary tumour ("-01") and each patient had exactly one sample, so the 295 mutant samples mapped to 295 patients.
5. **Merge with clinical data** → The patient-level TP53 table was left-joined onto the 566-patient clinical table, and patients with no TP53 record were labelled wildtype. This is only valid if every patient was sequenced; the study's case list (`cases_sequenced.txt`) confirms that all 566 samples have mutation data.

2.3 Classifying TP53 mutations

Raw mutation annotations (the MAF Variant_Classification field) were simplified as shown in Table 1. A patient was classed as TP53-mutant if they carried at least one protein-altering TP53 mutation. Silent mutations, which do not change the amino-acid sequence, were removed before this step (see Section 3.6).

Table 1. *TP53 mutation classification scheme*

Raw annotation (Variant_Classification)	Simplified class	Expected effect on p53	Records
Missense_Mutation	Missense	One amino acid changed; full-length protein	195

Nonsense_Mutation	Nonsense	Premature stop codon; truncated or no protein	55
Frame_Shift_Del / Frame_Shift_Ins	Frameshift	Reading frame disrupted; truncated protein	27
Splice_Site	Splice	Exon skipping or intron retention; usually loss of function	23
Splice_Region	Splice region (kept separate)	Near an exon–intron boundary; effect uncertain	9
In_Frame_Del / In_Frame_Ins	In-frame indel	Amino acids removed or added; frame intact	5
Silent	Excluded	Protein sequence unchanged	4

*Records = number of TP53 mutation records in data_mutations.txt (total 318).

Grouping decision: Following the project guide, I counted the patients in each category before choosing the final groups, and chose without looking at survival. The frameshift (23 patients) and splice (19 patients) groups were too small for stable survival estimates (Table 4), so patients were merged into four groups:

a. **Wildtype** → no protein-altering TP53 mutation.
b. **Missense** → missense mutation(s) only.
c. **Truncating/Splice** → nonsense, frameshift and/or splice-site mutation(s) only, all expected to cause loss of function.
d. **Other/mixed** → in-frame indels, splice-region variants only, or mutations from more than one class (for example: missense plus nonsense). This group (20 patients with survival data) was too small and too varied to interpret, so it was left out of the three-group comparison but kept in the "mutant" group for the wildtype-versus-mutant comparison.

2.4 Survival analysis

The outcome was **overall survival (OS)**: the time from diagnosis to death from any cause (OS_MONTHS). Patients who were alive at their last follow-up (OS_STATUS "0:LIVING")

were censored: we know they survived at least that long, but not how much longer. Key terms are explained in Appendix B.

1. **Kaplan–Meier (KM) curves** [15] → estimate the probability of surviving beyond each time point while making correct use of censored patients. Median OS and 3-year OS are reported with 95% confidence intervals (CI).
2. **Log-rank tests** [16] → test whether survival curves differ overall. Two comparisons were made: 3A wildtype vs mutant, and 3B wildtype vs missense vs truncating/splice (2 degrees of freedom), followed by pairwise tests. Pairwise tests were not corrected for multiple testing and are treated as exploratory.
3. **Cox proportional-hazards regression** [17] → estimates hazard ratios (HR). An HR of 1.5 means a 50% higher risk of death at any given time. Three models were fitted: (1) TP53 group only; (2) TP53 group + stage (the main model, *Survival ~ TP53 mutation type + Stage*); (3) TP53 group + stage + age + sex, as a check. Models 1 and 2 used exactly the same patients (those with known stage), so any change in the HR reflects the adjustment rather than a different sample.
4. **Model checks** → The proportional-hazards assumption was tested with Schoenfeld residuals [18]. Three sensitivity analyses tested whether the conclusions depended on analysis choices (Section 3.6). Statistical power was estimated with Schoenfeld's formula [19].
5. **Software** → Python 3.12 with pandas, NumPy, SciPy, matplotlib and lifelines 0.30 [20]. An earlier version of the analysis that implemented the KM estimator, log-rank test and Cox model directly in NumPy gave the same results (log-rank $p = 0.033$ and 0.177 under the original classification rules), which served as an independent cross-check. All tests were two-sided with significance set at $p < 0.05$.

3. Results

3.1 Study population

Of the 566 patients in the study, 61 had no survival data, leaving **505 patients** for the survival analyses, of whom 182 died during follow-up (Figure 1). Median follow-up, estimated with the reverse Kaplan–Meier method, was 28.7 months. Most tumours were early stage (54.5% stage I), reflecting the fact that TCGA collected tissue mainly from surgically removed tumours.

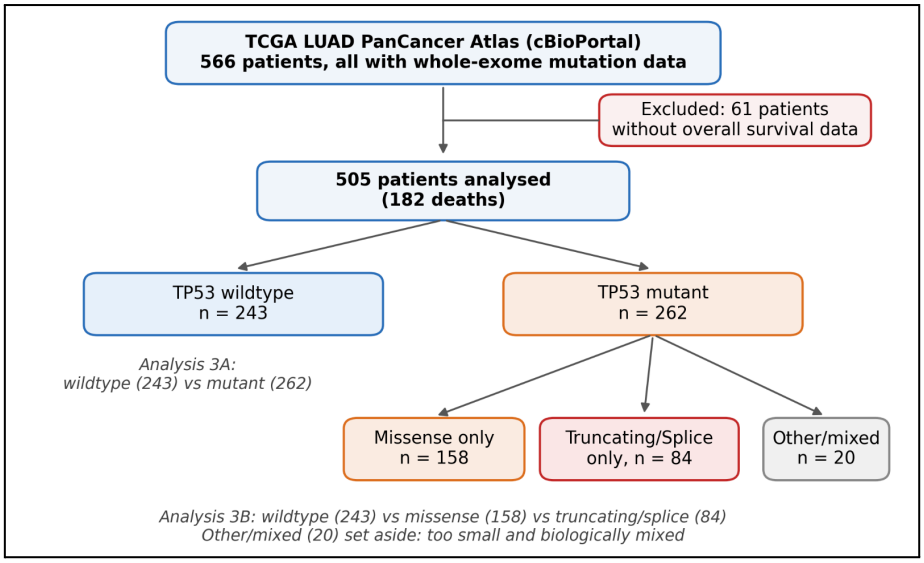


Figure 1. *Study Flow*

Table 2. *Characteristics of the 505 patients with survival data, by TP53 group.*

	All	Wildtype	Missense	Trunc./Splice	Other/mixed
Patients, n	505	243	158	84	20
Deaths, n (%)	182 (36%)	82 (34%)	54 (34%)	35 (42%)	11 (55%)
Age, median (IQR), years	66 (59-73)	68 (61-74)	64 (58-72)	62 (57-70)	70 (58-72)
Male, %	47	46	47	50	45
Stage I, n (%)	271 (55%)	135 (57%)	77 (49%)	48 (59%)	11 (55%)
Stage II, n (%)	120 (24%)	55 (23%)	45 (28%)	16 (20%)	4 (20%)
Stage III, n (%)	80 (16%)	37 (16%)	30 (19%)	8 (10%)	5 (25%)
Stage IV, n (%)	26 (5%)	11 (5%)	6 (4%)	9 (11%)	0 (0%)
Stage unknown, n	8	5	0	3	0

*Stage percentages use patients with known stage. Stage distribution across wildtype, missense and truncating/splice groups: χ^2 test $p = 0.058$. Age: Kruskal–Wallis $p < 0.001$ (wildtype patients were older). IQR = interquartile range.

The groups were similar in sex and broadly similar in stage (χ^2 $p = 0.058$), although missense patients had fewer stage I tumours than wildtype patients (49% vs 57%) and more stage II–III tumours (47% vs 39%). This difference turned out to matter for the stage-adjusted analysis (Section 3.5). Wildtype patients were older than mutant patients (median 68 vs 62–64 years).

3.2 Analysis 1: How common are TP53 mutations?

295 of 566 patients (52.1%) carried at least one protein-altering TP53 mutation (Figure 2A). At the record level, the 318 TP53 records were dominated by missense mutations (195; 61.3%), followed by nonsense (55; 17.3%), frameshift (27; 8.5%) and splice-site (23; 7.2%) mutations, with a few splice-region (9), in-frame (5) and silent (4) changes. At the patient level (Table 3), 178 patients (31.4%) had only missense mutations and 95 (16.8%) had only truncating/splice mutations. Nineteen patients carried two or more TP53 mutations, which is exactly why records and patients had to be counted separately.

Table 3. *Characteristics of the 505 patients with survival data, by TP53 group.*

TP53 category	Patients, n	% of 566	With survival data, n
Wildtype	271	47.9	243
Missense	178	31.4	158
Nonsense	53	9.4	46
Frameshift	23	4.1	21
Splice	19	3.4	17
Other/mixed	22	3.9	20
Total TP53-mutant	**295**	**52.1**	**262**

*Nonsense, frameshift and splice are later merged into the Truncating/Splice group. Other/mixed = in-frame indels, splice-region variants only, or mutations from more than one class.

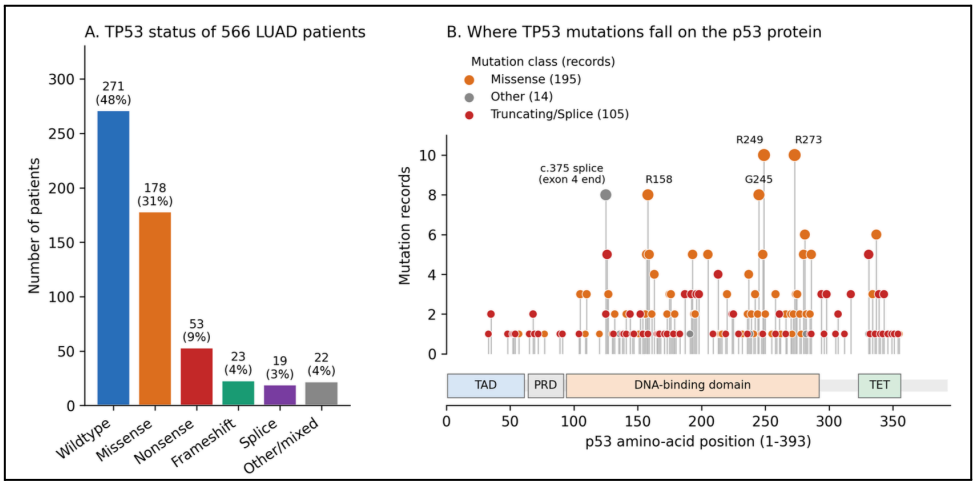


Figure 2. *The TP53 mutation landscape. (A) TP53 status of all 566 patients. (B) Position of each protein-altering mutation record along the 393-amino-acid p53 protein*

Mutations were not spread evenly along the protein (Figure 2B). 93% of missense mutations fell in the DNA-binding domain (residues 94–292), compared with 51% of nonsense and frameshift mutations, which were scattered along the whole protein. The most frequently hit residues were R249 and R273 (10 records each) and R158 and G245 (8 each), all well-known TP53 hotspots. The most common change at each of these (R158L, G245V, R249M and R273L) is a G>T transversion, and C>A/G>T changes made up 43% of all TP53 point mutations in this cohort: the characteristic mutational fingerprint of tobacco-smoke carcinogens [12].

3.3 Analysis 2: How common are TP53 mutations?

Table 4. *Overall survival by detailed TP53 mutation type (Analysis 2)*

TP53 group	n	Deaths	Median OS, months (95% CI)	3-year OS, % (95% CI)
Wildtype	243	82	53.7 (46.7–88.1)	68.3 (60.4–74.9)
Missense	158	54	44.6 (36.7–86.0)	62.1 (51.2–71.2)
Nonsense	46	15	56.7 (31.2–110.5)	60.8 (37.4–77.8)
Frameshift	21	12	24.0 (6.9–NR)	46.6 (24.4–66.1)
Splice	17	8	42.3 (20.2–47.8)	51.1 (19.4–76.0)
Other/mixed	20	11	24.2 (18.4–41.6)	24.0 (4.1–52.8)

Wildtype patients had a median OS of 53.7 months. Among the mutant subgroups the picture was inconsistent: patients with nonsense mutations (n = 46) had a median OS of 56.7 months, whereas patients with frameshift mutations (n = 21) had a median of only 24.0 months. The confidence intervals for the small groups were extremely wide (for example, 3-year OS for the splice group could be anywhere from 19% to 76%). This confirmed that the frameshift and splice groups were too small to analyse on their own and supported merging them with nonsense mutations into one truncating/splice group. The Other/mixed group had the poorest 3-year OS (24%), but with only 20 patients of several different kinds, this cannot be interpreted.

3.4 Analysis 3: Kaplan–Meier curves and log-rank tests

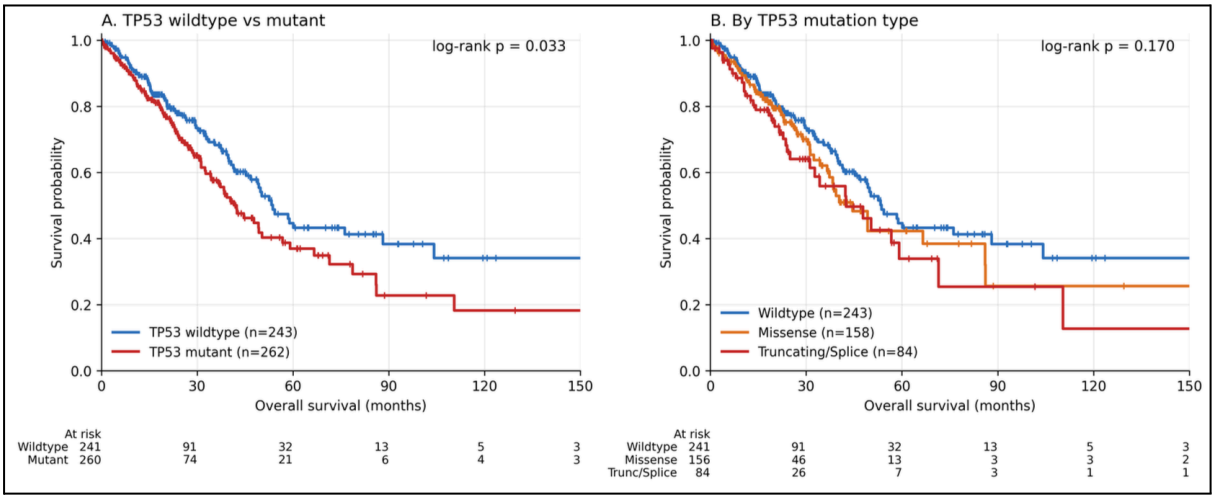


Figure 3. *Kaplan–Meier survival curves of LUAD patients stratified by TP53 status. (A) TP53 wildtype vs mutant. (B) Wildtype vs missense vs truncating/splice (Other/mixed excluded).*

Table 5. *Kaplan–Meier estimates and log-rank tests*

Group	n	Deaths	Median OS, months (95% CI)	3-year OS, % (95% CI)	Log-rank test
Wildtype	243	82	53.7 (46.7–88.1)	68.3 (60.4–74.9)	
Mutant (all)	262	100	42.3 (34.4–50.3)	57.7 (49.4–65.1)	3A: $\chi^2 = 4.57$, $p = 0.033$
Missense	158	54	44.6 (36.7–86.0)	62.1 (51.2–71.2)	3B: $\chi^2 = 3.55$, $p = 0.170$
Truncating/Splice	84	35	42.5 (31.2–71.5)	55.9 (41.4–68.2)	

- 3A: Wildtype vs Mutant

TP53-mutant patients had shorter survival than wildtype patients: median OS 42.3 vs 53.7 months and 3-year OS 57.7% vs 68.3% (log-rank p = 0.033). The curves separated within the first year and stayed apart (Figure 3A).

- 3B: By Mutation Type

The order of the groups matched my prediction: 3-year OS was 68.3% for wildtype, 62.1% for missense and 55.9% for truncating/splice, but the differences were not statistically significant (log-rank p = 0.170). The closest pairwise comparison was wildtype vs truncating/splice (p = 0.069), and missense and truncating/splice did not differ (p = 0.428).

The truncating/splice curve was lowest during the first three years, but the missense and truncating/splice curves crossed repeatedly after that, when very few patients were still being followed (13 and 7 at 60 months).

3.5 Analysis 4: Is the association independent of tumour stage?

Stage was a very strong predictor of survival (Figure 4A): median OS was 86.1 months for stage I, compared with 39.7, 26.5 and 27.2 months for stages II, III and IV (log-rank p < 0.001). In the Cox model, stages II–IV carried hazard ratios of 2.5–3.9 relative to stage I, which is far larger than any TP53 effect (Table 6, Figure 4B).

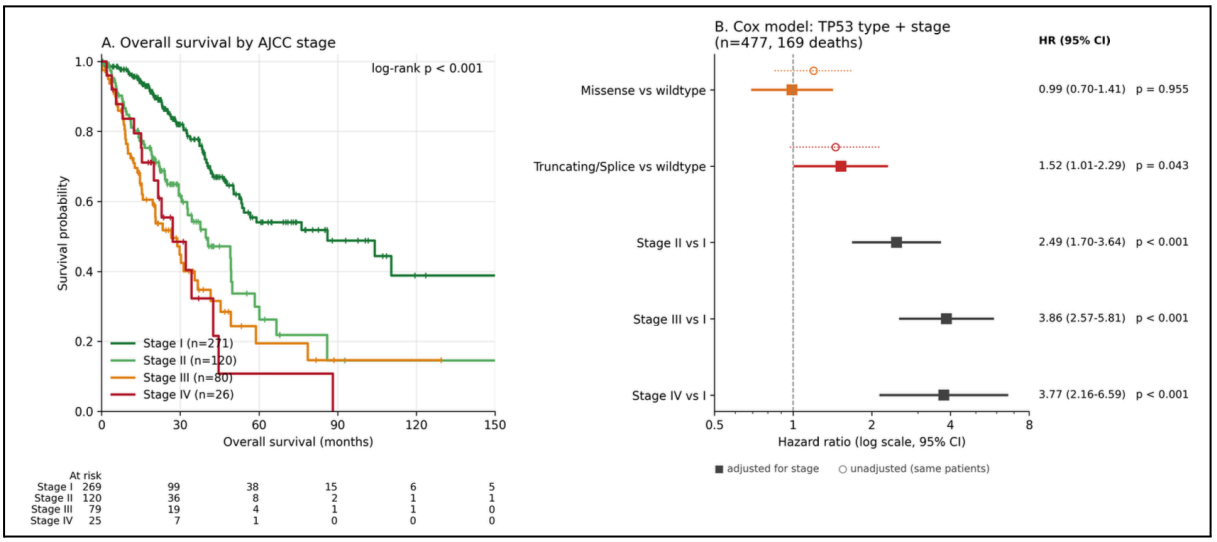


Figure 4. *(A) Overall survival by AJCC pathologic stage. (B) Hazard ratios from the stage-adjusted Cox model (filled squares) compared with unadjusted estimates in the same 477 patients (open circles).*

Table 6. *Cox proportional-hazards models: hazard ratio (95% CI), p-value*

Term	Model 1: unadjusted	Model 2: + stage	Model 3: + stage, age, sex
Comparison A: TP53 mutant vs wildtype	n = 497, 180 deaths	n = 497, 180 deaths	n = 487, 177 deaths
Mutant vs wildtype	1.37 (1.02–1.84), p = 0.037	1.21 (0.90–1.62), p = 0.217	1.25 (0.92–1.70), p = 0.156
Stage II vs I	—	2.35 (1.63–3.38), p < 0.001	2.22 (1.53–3.21), p < 0.001
Stage III vs I	—	3.50 (2.39–5.13), p <	3.23 (2.20–4.75), p < 0.001

		0.001	
Stage IV vs I	—	3.75 (2.16–6.52), p < 0.001	3.66 (2.09–6.42), p < 0.001
Comparison B: TP53 mutation type	n = 477, 169 deaths	n = 477, 169 deaths	n = 467, 166 deaths
Missense vs wildtype	1.20 (0.85–1.69), p = 0.305	0.99 (0.70–1.41), p = 0.955	1.04 (0.73–1.50), p = 0.813
Truncating/Splice vs wildtype	1.45 (0.97–2.17), p = 0.068	1.52 (1.01–2.29), p = 0.043	1.52 (1.01–2.30), p = 0.045
Stage II vs I	—	2.49 (1.70–3.64), p < 0.001	2.31 (1.58–3.40), p < 0.001
Stage III vs I	—	3.86 (2.57–5.81), p < 0.001	3.53 (2.34–5.31), p < 0.001
Stage IV vs I	—	3.77 (2.16–6.59), p < 0.001	3.68 (2.09–6.47), p < 0.001
Age (per 10 years)	—	—	1.14 (0.97–1.34), p = 0.119
Male vs Female	—	—	1.02 (0.75–1.39), p = 0.895

Wildtype vs mutant: Before adjustment, TP53-mutant patients had a 37% higher hazard of death (HR 1.37, 95% CI 1.02–1.84, p = 0.037). After adding stage, the hazard ratio fell to 1.21 (0.90–1.62) and was no longer statistically significant (p = 0.217).

By mutation type: The two mutation groups behaved differently once stage was added. The missense hazard ratio fell from 1.20 to 0.99 (p = 0.955), which is essentially no difference from wildtype. The truncating/splice hazard ratio did not fall; it rose slightly from 1.45 to 1.52 (95% CI 1.01–2.29, p = 0.043). Adding age and sex changed nothing (missense HR 1.04; truncating/splice HR 1.52, p = 0.045), and neither age nor sex was itself significant.

The explanation is visible in Table 2: missense patients more often had stage II–III tumours (47% vs 39% for wildtype), so part of their worse unadjusted survival reflected their stage. Truncating/splice patients were not diagnosed at more advanced stages overall (59% were stage I), so adjusting for stage did not explain away their poorer survival.

3.6 Checking the results: model assumptions and sensitivity analyses

Proportional hazards: The Cox model assumes that each hazard ratio stays constant over time. The Schoenfeld test showed no problem for any TP53 term (p = 0.35–0.44), but the stage III term did not meet the assumption (p = 0.011): its effect seems to change over time (in Figure 4A the stage III curve falls steeply at first and then levels off). To make sure this did not distort the TP53 estimates, I refitted the model stratified by stage, which gives each stage its own baseline risk instead of a single hazard ratio. The TP53 results were unchanged (S3 in Table 7).

Table 7. *Sensitivity analyses*

Analysis	Groups (WT / Mis / Trunc)	Log-rank p, 3 groups	Missense HR (p)*	Trunc./Splice HR (p)*
Main analysis	243/158/84	0.170	0.99 (0.955)	1.52 (0.043)
S1: original Phase 1 rules (silent variants counted as “Other”)	243/157/84	0.177	0.97 (0.887)	1.52 (0.043)
S2: exon-boundary splice-region variants counted as splice mutations	243/158/92	0.050	0.99 (0.953)	1.65 (0.010)
S3: main classification, Cox model stratified by stage	243/158/84	—	1.03 (0.850)	1.53 (0.041)

*Hazard ratio vs wildtype from the stage-adjusted Cox model. WT = wildtype; Mis = missense; Trunc = truncating/splice

Two data-quality discoveries:

1. Silent Mutations

My original Phase 1 script labelled the four silent TP53 variants as “Other”. Silent variants do not change the protein, and all four occurred in patients who also had a real TP53 mutation, so no one’s mutant/wildtype status was wrong. However, two missense and two nonsense patients were pushed into the “mixed” group and dropped from the mutation-type comparison. The pipeline now removes silent variants before grouping. With the original rules (S1), the results were almost identical (three-group p = 0.177 vs 0.170).

2. Splice-region Variants

Eight patients were classed as mutants only because of a “splice-region” variant. Seven of them carried a change at exactly the same nucleotide, c.375, the last base of exon 4, and the eighth at c.993, the last base of exon 9. Although annotated as “synonymous” (no amino-acid change), a change at the final base of an exon can destroy the splice donor site, and ClinVar classifies c.375G>A as pathogenic for this reason [21]. The same nucleotide being hit seven times in 566 tumours also suggests these variants were selected for during tumour growth. When they were counted as splice mutations (S2), the truncating/splice group grew from 84 to 92 patients, the three-group log-rank p-value fell to 0.050 and the adjusted hazard ratio rose to 1.65 ($p = 0.010$). Because I made this reclassification after seeing the data, I report it only as a sensitivity analysis. Its main lesson is how much a handful of classification decisions can move a borderline result.

4. Discussion

4.1 Overall Discoveries

I expected truncating/splice mutations to carry the worst prognosis, and the type of mutation to make a clear difference. What I found was more complicated. TP53-mutant patients as a whole did have shorter survival ($p = 0.033$), but the three mutation groups were not significantly different from one another ($p = 0.170$), and the missense and truncating/splice curves overlapped heavily.

The unexpected part came from the stage-adjusted analysis. Once stage was included, the overall TP53 association largely disappeared (HR 1.37 → 1.21) and the missense association vanished completely (HR 0.99). Only the truncating/splice group remained associated with worse survival (HR 1.52, $p = 0.043$). So the 10 direction of my prediction held up, but the evidence is weak: the p-value is only just below 0.05, it is one of more than a dozen tests in this report, the lower end of its confidence interval is almost exactly 1.0, and the direct comparison of missense with truncating/splice was not significant ($p = 0.43$). I therefore interpret the result as consistent with the hypothesis, but not proof of it. The project guide posed five questions for interpreting a result like this; I address each below:

4.2 Was the sample large enough?

Not really. In survival analysis, statistical power depends on the number of deaths, not the number of patients. The truncating/splice vs wildtype comparison had 117 deaths. Using Schoenfeld's formula [19], detecting a hazard ratio of 1.5 with 80% power would need about 250 deaths; with 117, the power was only about 48%. For the smaller unadjusted wildtype-vs-mutant effect (HR 1.37), power was about 32%, and about 415 deaths would be needed. A non-significant result in this study therefore cannot be read as "mutation type does not matter": absence of evidence is not evidence of absence. Follow-up was also short (median 28.7 months), and very few patients were still being followed after five years, so the later parts of the curves are unreliable.

4.3 Is "mutation class" too coarse?

Probably. Even within the truncating group, patients with nonsense and frameshift mutations had very different median survival (56.7 vs 24.0 months; Table 4), although both groups were small. Missense mutations are not all alike either: some hotspot mutants are thought to act as dominant-negatives or to gain new functions, whereas others simply lose function [8, 22]. Putting all missense mutations in one group may average out effects that point in opposite directions. The splice-region finding (Section 3.6) shows a related problem: annotation categories do not always match biology. Classes based on measured protein function [22], or on hotspot versus non-hotspot position, might separate patients more meaningfully.

4.4 Do different mutations affect p53 function differently?

Figure 2B suggests they do. Missense mutations clustered almost entirely (93%) in the DNA-binding domain, producing a full-length protein that cannot bind its target genes. Because p53 works as a group of four (a tetramer), such a mutant can join with normal p53 copies and disable them. Truncating mutations instead remove the protein, often because the faulty mRNA is destroyed before it is translated. On this reasoning, one might even expect missense mutations to be worse than truncation, which is the opposite of my hypothesis, and both views appear in the literature, which is part of what makes the question interesting. In many tumours, however, the second, normal copy of TP53 is also lost (loss of heterozygosity), and once no normal p53 remains, the difference between a dominant-negative and a simple loss-of-function mutation becomes smaller. Nineteen patients

here carried two TP53 mutations, possibly one on each copy. My dataset did not include copy-number data, so I could not test this.

4.5 Is stage a stronger prognostic factor?

Clearly yes. Stage II–IV tumours carried hazard ratios of 2.5–3.9, compared with at most about 1.5 for any TP53 group. Stage can act as a confounder: it affects survival, and if it is also linked to TP53 status, a crude comparison mixes the two effects together. That appears to be what happened with missense mutations. There is a subtlety, though. If TP53 mutations make tumours grow and spread faster, they could cause later-stage diagnosis; stage would then be a mediator on the path from mutation to death, and adjusting for it would hide part of a real TP53 effect. Observational data like these cannot fully separate the two explanations, so the adjusted and unadjusted results are best read together.

4.6 Could other co-occurring mutations explain the results?

Possibly. Lung adenocarcinomas carry other driver mutations, including KRAS, EGFR, STK11 and KEAP1 [1, 14]. Some of these are linked to TP53 status and to survival: EGFR-mutant tumours can be treated with targeted drugs, while STK11 and KEAP1 mutations have been associated with poor outcomes [14]. If, for example, truncating TP53 mutations happened to co-occur more often with an unfavourable partner mutation, that partner (not TP53) could explain the difference. Smoking history, reflected in the G>T mutation pattern seen in Figure 2B, is another factor linked both to TP53 mutations and to survival. I deliberately kept the model to TP53 type and stage, but co-mutations are the most important next variable to examine.

5. Limitations

1. **Observational design**: This is a retrospective analysis of existing data, so it can show associations but not cause and effect.
2. **Who is in TCGA:** TCGA tissue came mainly from surgically removed tumours (54.5% stage I), so the results may not apply to advanced lung cancer. Treatments (chemotherapy, targeted therapy, immunotherapy) were not modelled.
3. **Outcome:** Overall survival counts deaths from any cause, including causes unrelated to cancer. Disease-specific survival or progression-free interval (both available in TCGA-CDR [3]) would be more specific to the tumour.

4. **Small groups and short follow-up:** limited statistical power (Section 4.2).
5. **Multiple testing:** More than a dozen tests were run. At a 0.05 threshold, about one "significant" result could be expected by chance alone, so the borderline truncating/splice result should be treated with caution.
6. **Model assumptions:** The stage III term violated the proportional-hazards assumption; this was addressed with a stratified model, which gave the same TP53 results.
7. **Classification by annotation only:** Mutations were classified from their database annotation, without functional or protein-level data, copy-number (second-allele) status or information on how many tumour cells carried each mutation.
8. **Missing data:** 61 patients lacked survival data, 8 lacked stage and 10 lacked age; these patients were excluded from the analyses that needed those variables.

6. Future directions

1. **Validate in an independent cohort**: such as the larger MSK lung adenocarcinoma cohorts in cBioPortal [23], using exactly the same classification rules, fixed in advance.
2. **Use finer, function-based classes:** hotspot vs non-hotspot missense mutations, or classes based on large-scale functional measurements of each TP53 variant [22].
3. **Add co-mutations one at a time** (KRAS, EGFR, STK11, KEAP1) to see whether they explain the truncating/splice association.
4. **Use copy-number data** to identify tumours that have lost both copies of TP53.
5. **Try other endpoints** such as disease-specific survival and progression-free interval from TCGA-CDR.

7. Conclusion

In 505 patients with lung adenocarcinoma, TP53 mutation was associated with shorter overall survival, but this association weakened substantially once tumour stage was taken into account. The type of mutation did not clearly separate patients: only truncating/splice mutations remained (borderline) associated with worse survival after stage adjustment, and the study did not have enough statistical power to settle the question. The main lessons are that tumour stage dominates prognosis, that "TP53-mutant" is too broad a label, and that

classification decisions affecting even a handful of patients can move a borderline result. A larger cohort with function-based mutation classes would be needed to test the hypothesis properly.

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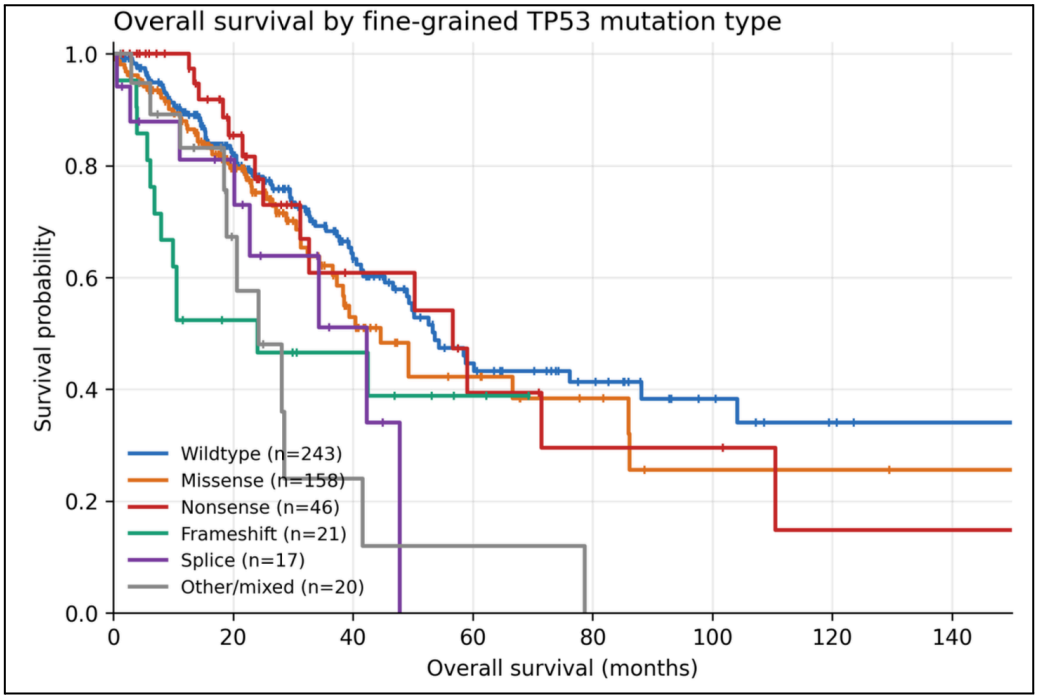
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Appendix A: Supplementary figure



Supplementary Figure S1. Kaplan–Meier curves for each detailed TP53 mutation type (Analysis 2). The small frameshift (n = 21), splice (n = 17) and Other/mixed (n = 20) curves drop in large, erratic steps because each death removes 5–6% of the group, which is why these groups were merged before formal testing.

Appendix B: Glossary

Table C1. Key terms used in this report

Term	Meaning
Overall survival (OS)	Time from diagnosis until death from any cause.
Censoring	A patient who was still alive at last contact. We know they survived at least that long, so they count as "at risk" until then but are never counted as a death.
Kaplan–Meier curve	A step-shaped curve showing the estimated proportion of patients still alive at each time. It drops at every death and correctly accounts for censored patients.
Median OS	The time at which the Kaplan–Meier curve crosses 50%: half the patients are estimated to have died by then.
Log-rank test	Compares whole survival curves by asking, at every death, whether

	deaths occurred in each group as often as expected if the groups had identical survival.
Hazard ratio (HR)	From a Cox model: how much higher (HR > 1) or lower (HR < 1) the risk of death is in one group than in the reference group, at any given moment.
95% confidence interval	The range of values compatible with the data. If it includes 1.0, the hazard ratio is not statistically significant at $p < 0.05$.
Confounder	A variable linked to both the exposure (TP53 type) and the outcome (survival) that can create or hide an association, e.g. tumour stage.
Statistical power	The probability that a study detects a real effect of a given size. Low power makes "not significant" results hard to interpret.
Missense/nonsense/ frameshift/splice	One amino acid changed / premature stop codon/reading frame shifted by an insertion or deletion / RNA splicing disrupted.

Appendix C: Code and data availability

All code, processed data, figures and result tables are in the project's GitHub repository (`LUAD-TP53-survival`). The raw cBioPortal files are not redistributed; the repository explains how to download them. The analysis runs in three steps:

Script	What it does	Main outputs
src/01_build_phase1_table.py	Phase 1: load data, extract TP53, link samples to patients, merge with clinical data, QC checklist	phase1_final_table.csv
src/02_mutation_landscape.py	Analysis 1: mutation frequencies, protein map, study flow diagram	Figures 1–2
src/03_survival_analysis.py	Analyses 2–4: KM, log-rank, Cox models, assumption checks, sensitivity analyses	Figures 3–4, S1; Tables 2, 4–7